

SHUBHAM PAWAR

Full Stack Engineer | Software Engineer

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Professional Summary

Full Stack Software Engineer with **3+ years** delivering production web platforms and distributed backend services for enterprise, government, and AI product teams. Works end to end in **Python (FastAPI)**, **Java (Spring Boot)**, **React.js / Next.js**, **SQL**, **Docker**, and **AWS** — covering REST API design, database performance tuning, secure authentication and authorization, containerized execution, CI/CD, and cloud deployment. Additionally experienced in **ML training infrastructure** and **reinforcement learning environments** supporting **SFT**, **RLHF**, and **RLVR** workflows. Comfortable owning a system from ambiguous requirement to measured production outcome.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL, Bash, PHP

Backend and APIs: FastAPI, Spring Boot, Node.js, Express.js, Laravel, RESTful APIs, Microservices, Asynchronous Programming, JWT, OAuth 2.0, RBAC, Spring Security

Frontend: React.js, Next.js, HTML5, CSS3, Tailwind CSS, AJAX, Responsive UI, Real-Time Dashboards

Databases: PostgreSQL, MySQL, Query Optimization, Indexing, ACID Transactions, Data Modeling

Cloud and DevOps: AWS (EC2, S3, Lambda, IAM, Bedrock), GCP, Docker, Docker Compose, Docker Buildx, CI/CD, Linux, Shell Scripting, Git, GitHub, GitLab

ML and AI Engineering: Reinforcement Learning Environments, SFT (Supervised Fine-Tuning), RLHF, RLVR (Reinforcement Learning with Verifiable Rewards), Reward Function Design, Training Data Pipelines, Large Language Models (LLM), AI Agents, Prompt Engineering, LiteLLM, Model Evaluation and Benchmarking

Practices and Tools: System Design, Agile, Scrum, TDD, Unit and Integration Testing, pytest, Playwright, Postman, Code Review, Application Security (XSS, CSRF, Secure Tokens)

Professional Experience

Ethara AI

Mar 2026 – Present

Full Stack Engineer

Gurugram, India

- Own end-to-end delivery of a distributed **AI training and evaluation platform** — Python backend services, **React.js / Next.js** dashboards, containerized execution, and **AWS** deployment — used daily by engineering and research teams.
- Engineer **reinforcement learning (RL) environments**: deterministic, sandboxed task environments with programmatic reward functions, automated verification, and reproducible resets, enabling **RLVR** training runs at scale.
- Build data and orchestration pipelines producing curated trajectories and preference pairs for **SFT** and **RLHF** post-training workflows, with schema validation, deduplication, and quality filtering for auditable datasets.
- Raised autonomous task completion to **80%+ pass rates** on benchmark repositories by designing a staged multi-model agent workflow with structured context injection.
- Reduced inference spend and surfaced previously silent pipeline failures by introducing response caching, token cost analytics, and live model API health probing across a multi-provider stack.
- Own **Docker** containerization, **CI/CD**, **AWS** deployment, and **JWT/RBAC** security for high-throughput workloads; partner with product and research stakeholders in **Agile** sprints to turn ambiguous requirements into shipped, measurable features.

Austere Systems Limited

Sep 2023 – Feb 2026

Software Developer

Pune, India

- Delivered four production platforms end to end — **ERP**, **Fund Disbursement Management System** (Haryana Government), **Audit Management SaaS**, and **Voice Broadcast** — owning backend, frontend, and deployment on each.
- Digitized manual, paper-driven government fund disbursement and multi-department audit processes into workflow-driven systems with enforced approval hierarchies and auditable transaction trails.

- Cut key report and listing query response times by roughly **40%** through indexing strategy, query rewrites, and transaction tuning on a live production database.
- Built secure, role-aware experiences across **Java Spring Boot**, **Python FastAPI**, and **React.js / Next.js**, serving multi-level user hierarchies for government and enterprise clients.
- Acted as **tech lead** for ERP product development — task breakdown, code reviews, and release readiness across a 12–15 person project team — and mentored junior developers.
- Deployed and maintained services on **AWS** and **GCP** with Git-based **CI/CD** workflows.

Key Projects

RL Environment and Post-Training Data Platform | *Python, Docker, RLVR, SFT, RLHF, AWS* **Mar 2026 – Present**

- Built containerized **RL training environments** exposing a uniform reset/step/score interface over real software repositories, so a model's actions are executed in isolation and scored automatically rather than by human review.
- Designed **programmatic reward functions** combining test-suite outcomes, lint diagnostics, and runtime signals into a single verifiable score — the reward backbone for **RLVR** runs.
- Implemented rollout collection and dataset generation for **SFT** and **RLHF**: trajectory capture, schema validation, deduplication, quality filtering, and preference-pair construction from ranked candidate solutions.
- Guaranteed reproducibility with pinned images, seeded execution, an inactivity watchdog, and arm64/amd64 auto-detection, so the same task yields the same reward across machines and training runs.

Kaiju — Agentic Code Generation Pipeline and Orchestrator | *Python, AWS Bedrock, Docker* **Mar 2026 – July 2026**

- Architected a 3-stage agent workflow (Draft, Lint Refine, Test Refine) over **AST-stubbed** Python code, where each stage receives a different slice of context and refines the previous stage's output.
- Abstracted **5+ models** behind **LiteLLM** — Claude, Kimi K2, GLM, and Minimax on **AWS Bedrock** plus **OpenAI GPT** — enabling like-for-like quality, latency, and cost benchmarking from one interface.
- Wrote an **AST-based Python stubber** preserving imports, decorators, and import-time calls, plus a CSV-driven batch orchestrator chaining fork, clone, stub, push, dataset creation, Docker build, and test-ID generation.
- Parallelized isolated execution via **Docker Buildx** and **ThreadPoolExecutor** with secure **AWS** credential flows, and shipped **FastAPI** services with **React.js / Next.js** dashboards exposing per-stage pass rate, cost, and latency in place of manual log inspection.

Enterprise SaaS Audit Management Platform | *React.js, REST APIs, Web Security* **Aug 2025 – Feb 2026**

- Implemented **role-based UI rendering for 7 user roles** with strict access control, secure authentication flows, and session-safe routing.
- Built real-time dashboards with SLA timers, audit status tracking, and offline sync indicators for field auditors on unreliable connections.
- Hardened the client against **XSS** and **CSRF** with escaped rendering, safe request patterns, and secure token handling.

Fund Disbursement Management System — Haryana Government | *Java, Spring Boot, MySQL* **Jan 2024 – Jun 2024**

- Developed secure **RESTful APIs** in **Java Spring Boot** for fund allocation, approval workflows, and transaction processing across multiple administrative levels.
- Enforced **Spring Security** and **JWT** role-based authorization spanning Admin, Officer, and Employee hierarchies with distinct approval authority.
- Guaranteed **ACID-compliant** transactions and data integrity through layered validation, tuned queries, and structured exception handling.

Education

Savitribai Phule Pune University

Aug 2019 – Aug 2023

Bachelor of Engineering in Computer Science — CGPA: 8.81 / 10

Pune, India